

ADNOC OFFSHORE SOUR GAS TRANSMISSION - ANNEX H QUALITY PLAN

Pipe Description: 42" (1067 mm) x 20.60 mm, Grade X65MS PSL 2, SAWH Sour Service | Governing Standard: ADNOC DGS-1300-001 / API 5L Annex H (NACE MR0175 / ISO 15156)

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.06%, P <= 0.010%, S <= 0.0010%, Ca/S: 1.5-3.0, CE_Pcm <= 0.18	ADNOC DGS / Annex H	H/W
2	HIC Test (Hydrogen Induced Cracking)	1 set (3 specimens) per heat / 3 heats	Pipe body, weld & HAZ	NACE TM0284 Sol A	CLR <= 10%, CTR <= 3.0%, CSR <= 1.0% in Solution A (pH 2.7)	API 5L Ek H.7	H/W
3	SSC Test (Sulfide Stress Cracking)	1 set (3 specimens) per qualification heat	Four-point bend or tensile	NACE TM0177 Method A	No failure at 90% AYS after 720 hours in 1 bar H2S	API 5L Ek H.7	H/W
4	Pipe Body & Weld Tensile Test	Once per test unit	Transverse body & weld	ASTM A370	Rt0.5: 450-600 MPa, Rm: 535-760 MPa, Y/T <= 0.90	API 5L Tablo H.1	H/W
5	CVN Charpy Impact Test (-20 °C)	1 set body + 1 set weld at -20 °C	Transverse Charpy V	ASTM A370	At -20 °C: Min Average 60 Joules, Min Individual 45 Joules	ADNOC DGS	H/W
6	Hardness Testing (100% Survey)	Once per test unit (16-point survey)	Cross section macro	ASTM E384	MAXIMUM 250 HV10 (Body, HAZ, Weld Cap & Root - Sour Limit)	NACE MR0175 / API 5L Ek H	H/W
7	Mill Hydrostatic Pressure Test	Each pipe (100% all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 100% SMYS (185 bar), Duration: MINIMUM 15 SECONDS	ADNOC DGS	H/W
8	Weld Seam & Plate Full NDT	100% automated UT + 100% plate scan	Full weld & plate body	ISO 10893-11/9	100% automated UT + 100% laminar scan (Class E1); no defect > 4 mm	API 5L Ek H & E	H/W
9	Pipe Ends UT Laminar Testing	100% of pipe ends (100 mm band)	Both pipe ends	API 5L Annex E.8	100 mm band on pipe ends; no defect > 4.0 mm	API 5L Ek E.8	W
10	Dimensional, Wall Thickness & Marking	100% of all pipes	Pipe body & ends	API 5L 10.2.8	Diameter: +-0.5%; Wall thickness: -5.0% / +12.5%; Marking with suffix "MS"	ADNOC DGS	W